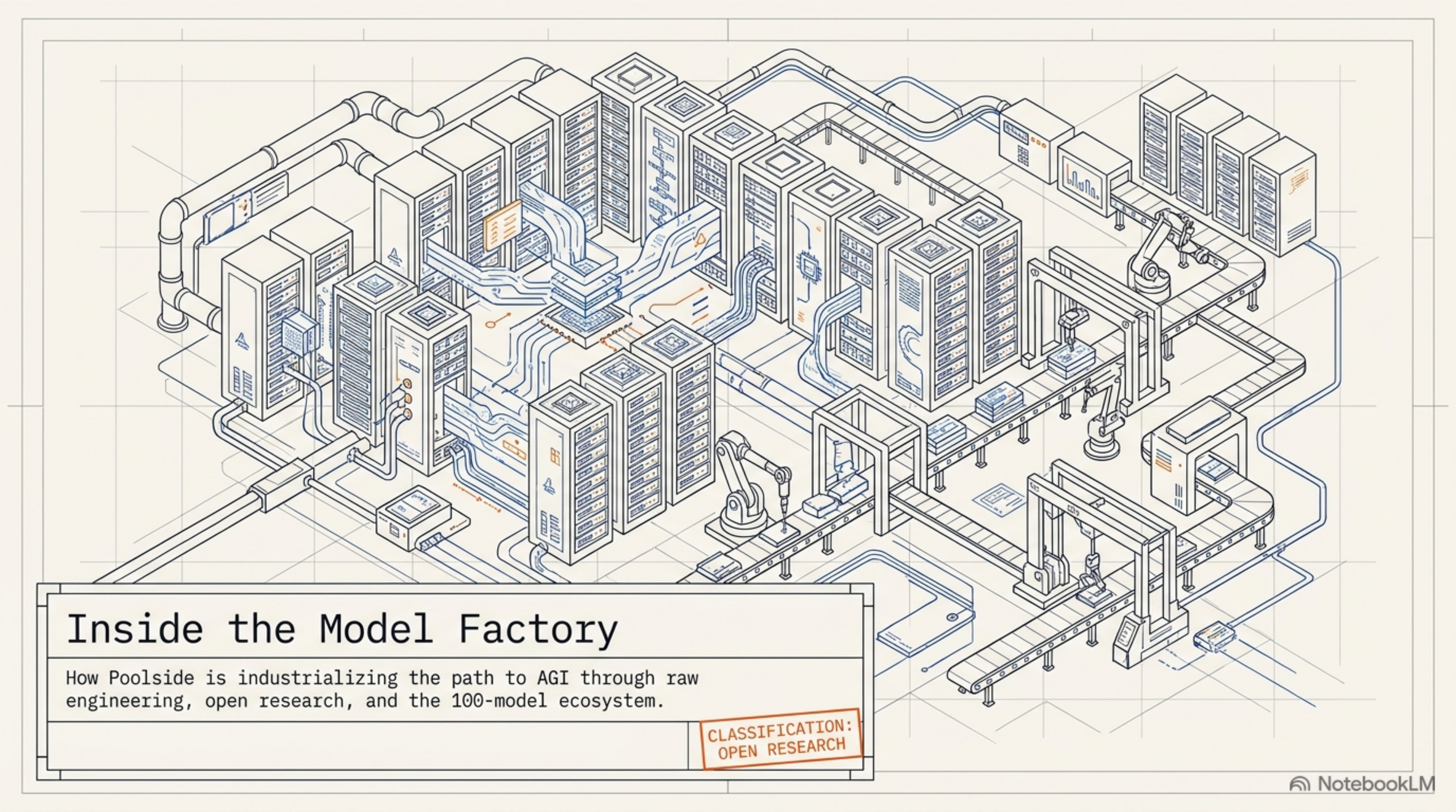


Inside the Model Factory

How Poolside is industrializing the path to AGI through raw engineering, open research, and the 100-model ecosystem.

CLASSIFICATION: OPEN RESEARCH

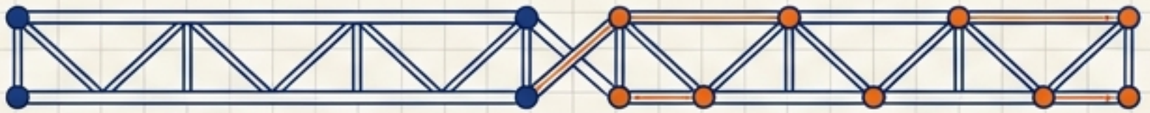
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A Fork in the Road for Foundation Models

The Big Tech Oligopoly	The Poolside Neo-Lab
A closed market controlled by 3 to 5 mega-corps.	A **100-company** ecosystem of choice and **open weights**.
Unilateral safety regulations and massive compute hoarding.	**Relentless** engineering execution and **open research**.
The hyper-localized Bay Area talent war.	A **globally distributed**, **high-agency** engineering force.
AI as a pure physics and math problem.	AI as a **90% software engineering** problem.

DIAGNOSTIC TABLE: SYSTEM ARCHITECTURE & STRATEGY // REV 1.4.B

SKUNKWORNS ENGINEERING LAB - LEAKED DOCUMENT



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The 90% Engineering Equation

95% [Data + Compute Efficiency]

Model building is an industrialized process. Success dictates relentlessly optimizing distributed systems, writing flawless code, and curating massive data pipelines.

+ 5% [Novel Architecture]

Pure scientific breakthroughs and novel attention mechanisms are critical, but they ultimately serve to unlock greater compute efficiency.

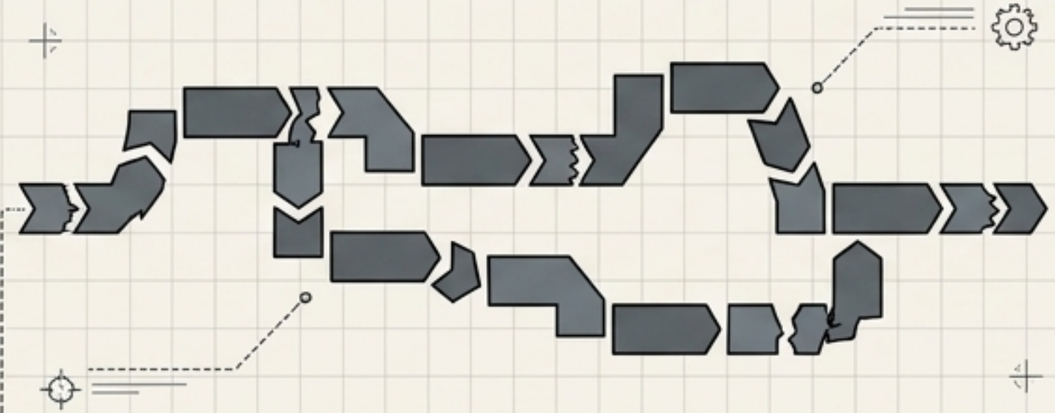
= Frontier Intelligence

A $12M failure in 2015 taught us that scaling and engineering persistence always trumps isolated scientific novelty.

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Rebuilding the Factory Floor

Legacy Training

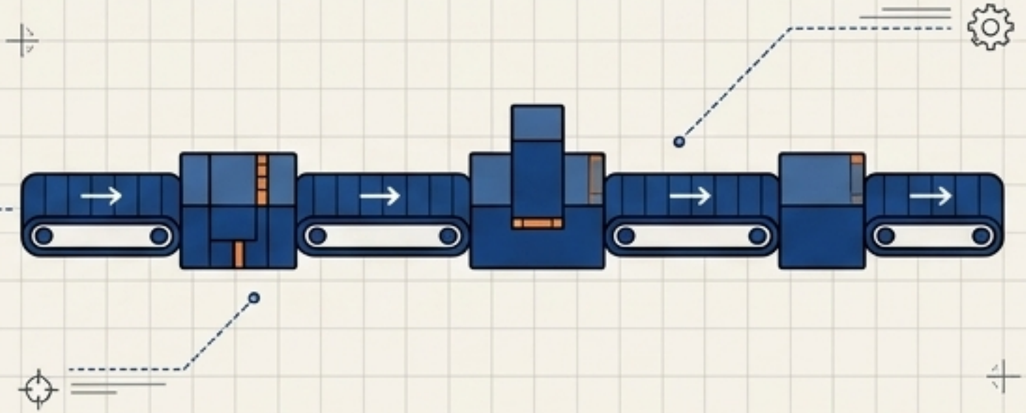


Process: Patchwork bash scripts & Slurm -> Repackaging massive datasets -> YOLO experimental runs.

Cycle Time: 6 months from idea to release.

Reliability: Constant human intervention and fragile on-call events.

The Model Factory



Process: Immutable data layers -> Version-controlled experiments -> Reproducible scientific rigor.

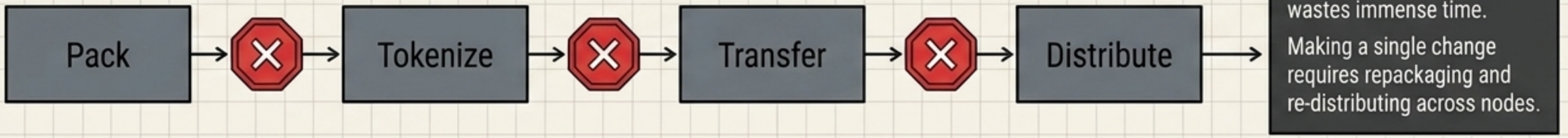
Cycle Time: 5 to 8 weeks from cold start to launch.

Reliability: Zero major on-call events. Fully automated pipelines.

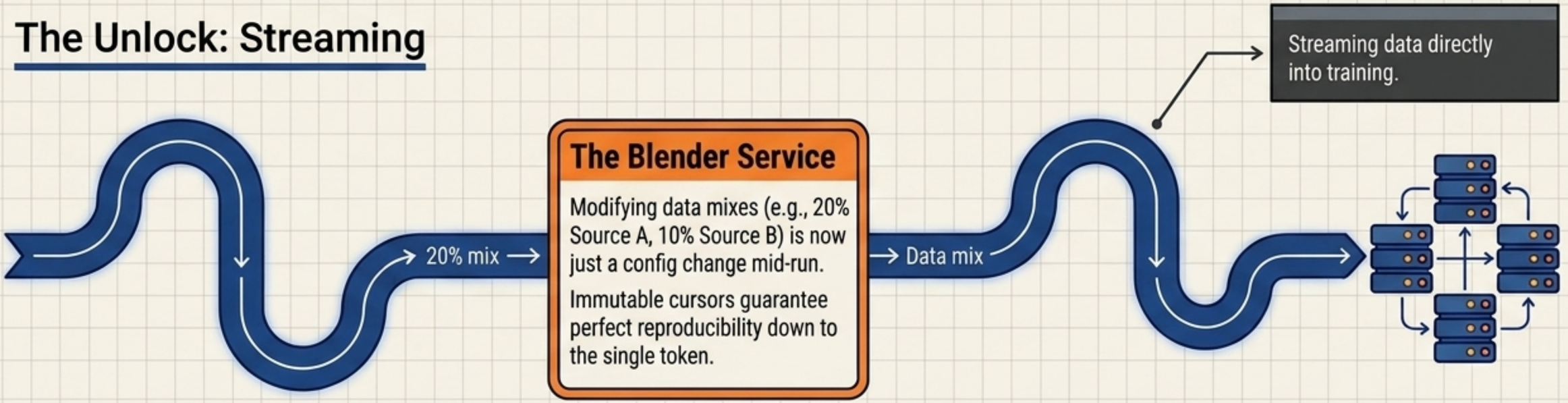
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The Just-in-Time Data Stream

The Old Way

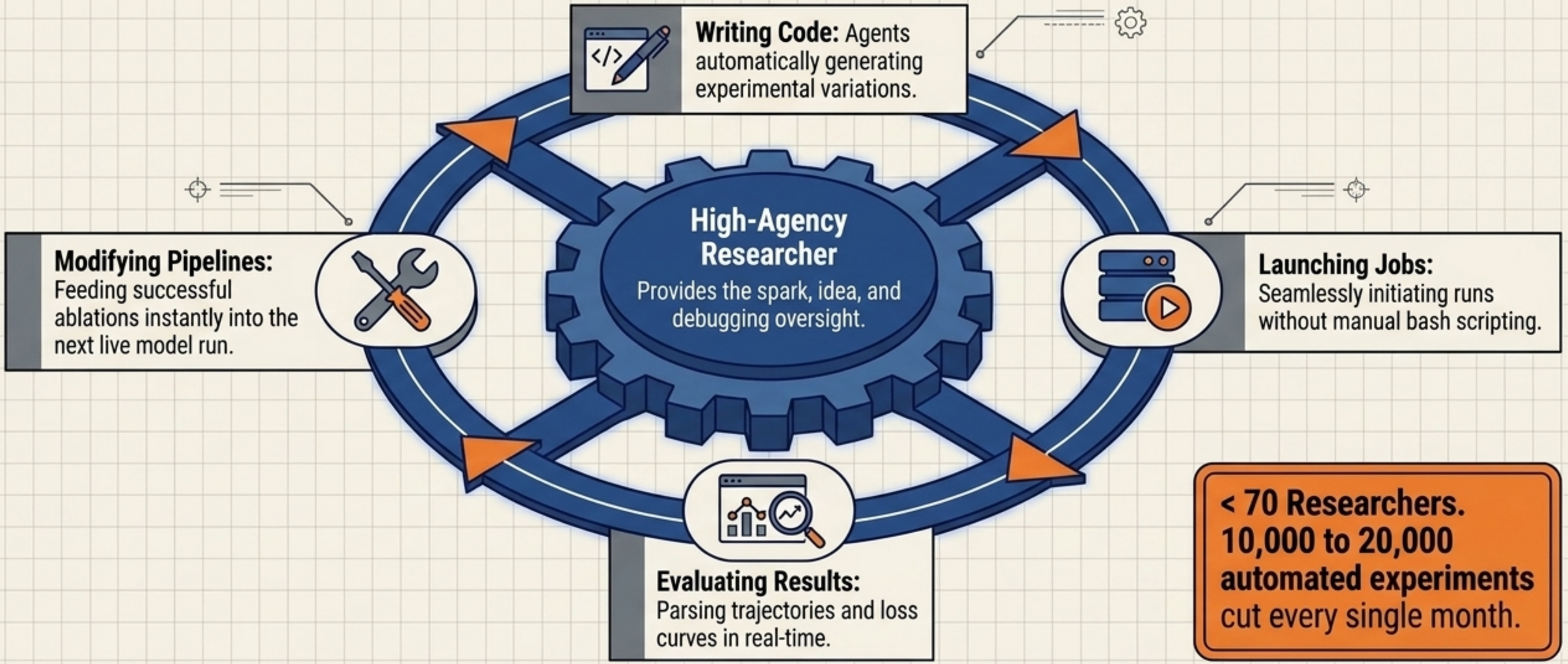


The Unlock: Streaming



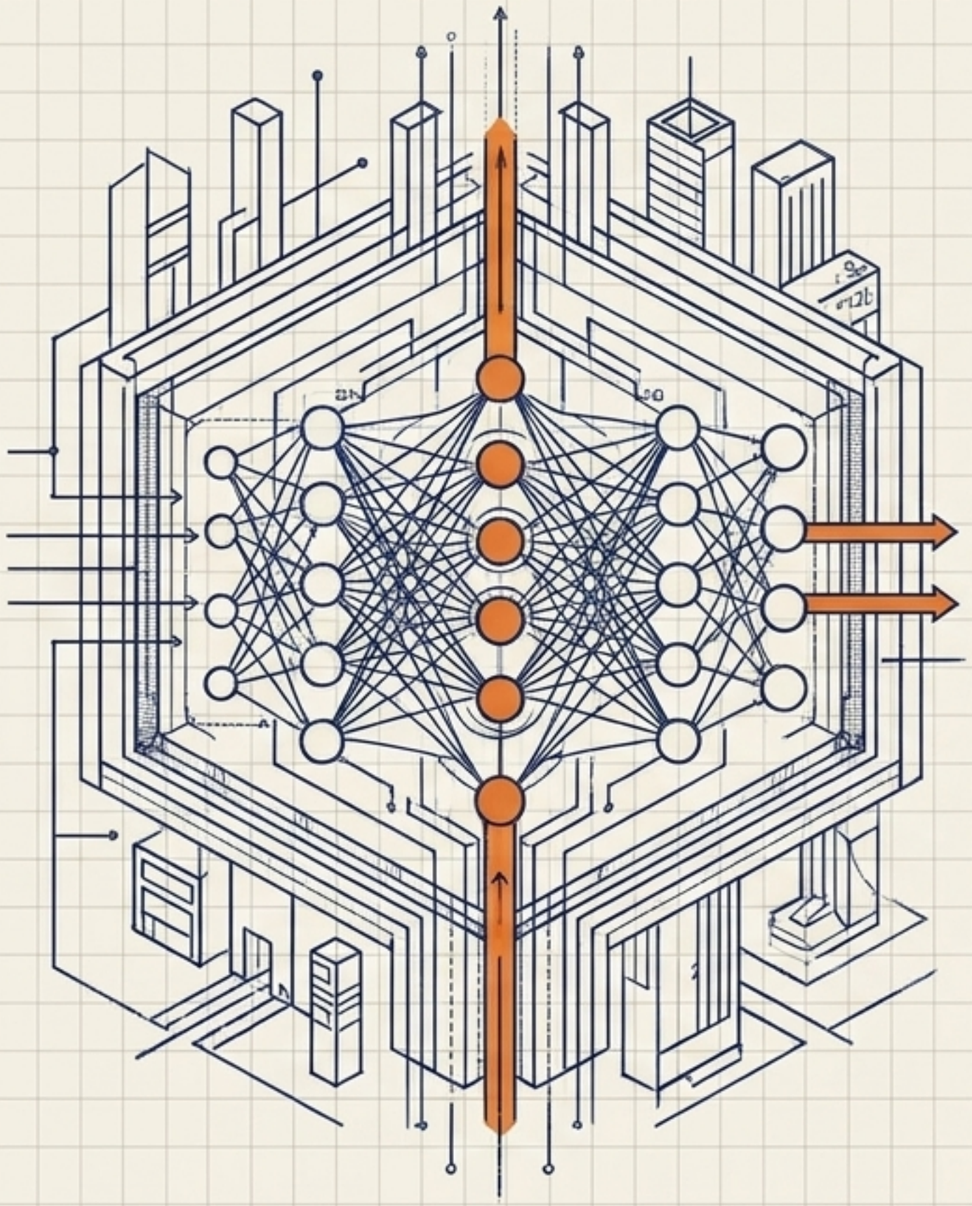
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Autonomous RSI on the Factory Floor



< 70 Researchers. 10,000 to 20,000 automated experiments cut every single month.

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The Output: Laguna S 2.1

Scale:
118 Billion
Total Parameters.

Sparsity:
8 Billion Active per token
(Mixture-of-Experts).

Context:
1 Million Token
Context Window.

Velocity:
8 weeks fmeriatores
from scratch to delivery.

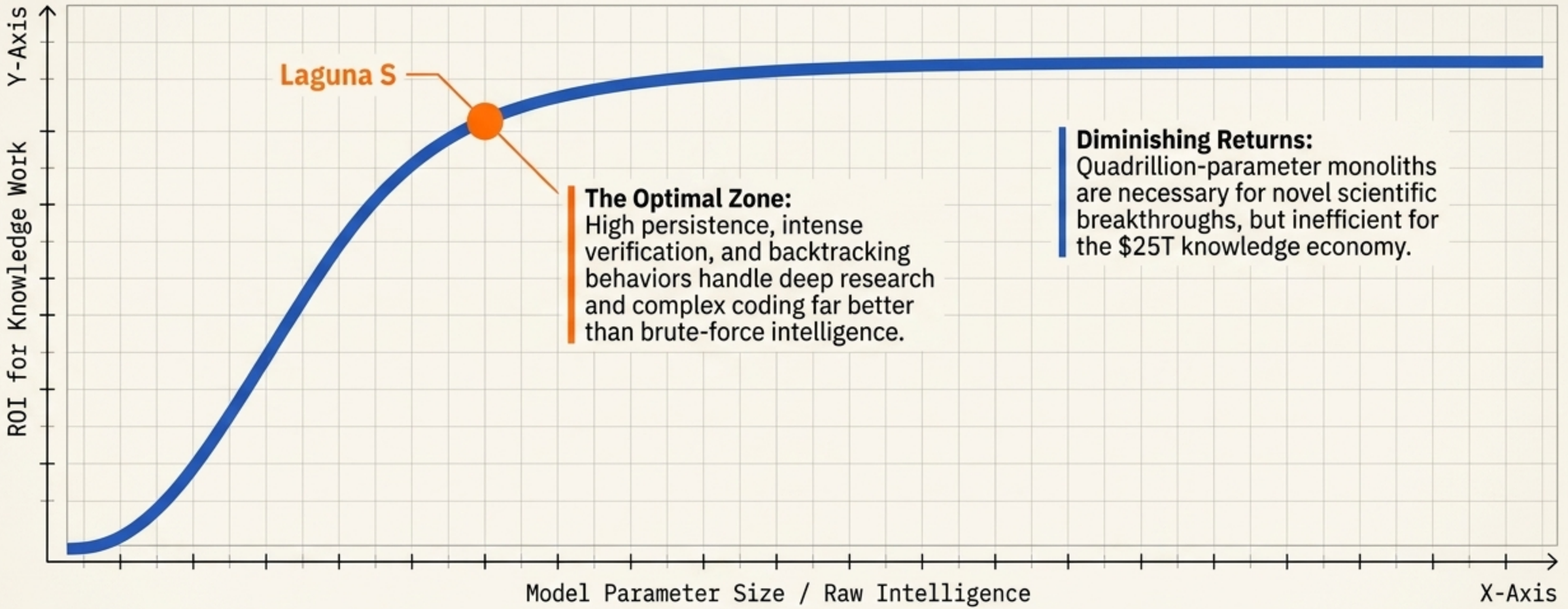
Velocity:
8 weeks from scratch to
delivery.

Hardware Fit:
Fits entirely on a single
DGX Spark.

By **leveraging the Model Factory**, this "small" weight class outperforms models nearly **10x** its size, proving optimization beats sheer scale.

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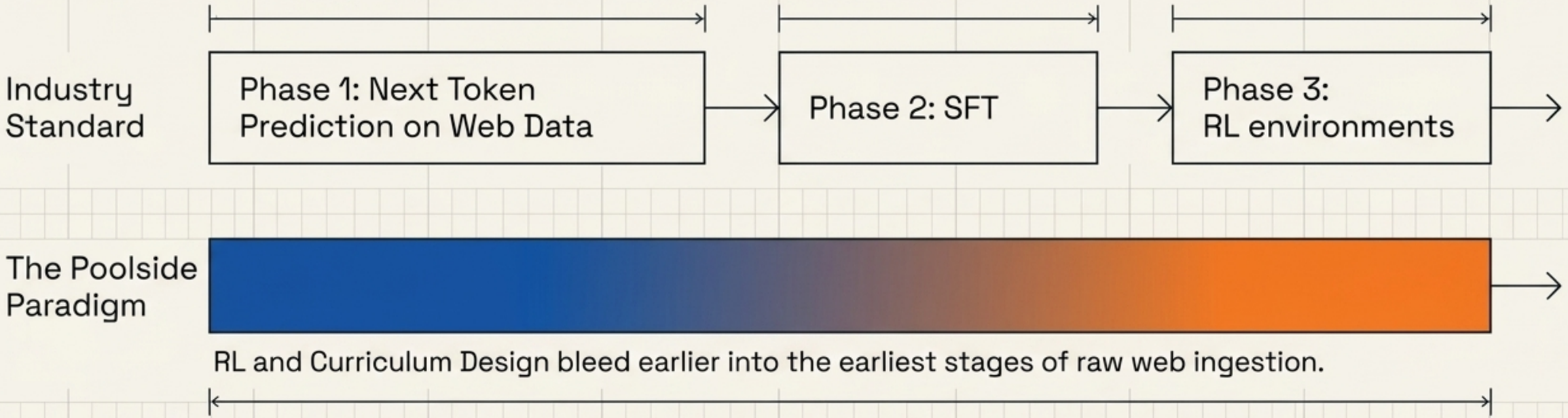
The Intelligence Curve: Persistence Over Raw IQ



We are squeezing far more capability out of smaller models by training for behavioral tenacity rather than raw parameter count.

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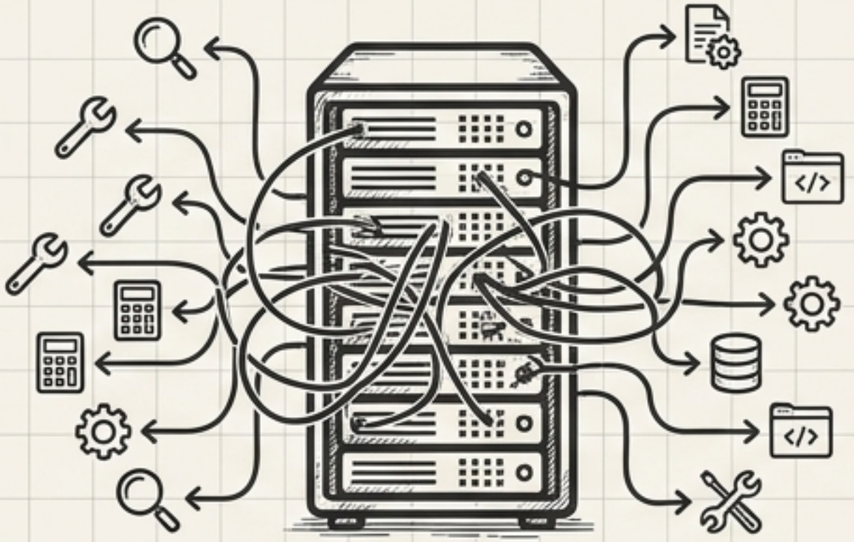
Reinventing the Curriculum

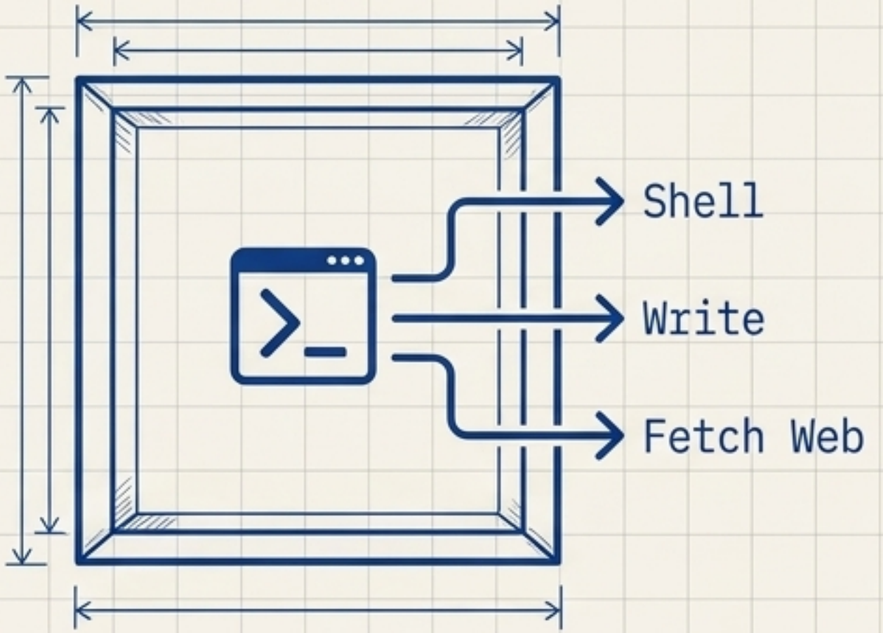


Next-token prediction is failing to extract enough knowledge from the web. The future is inducing reasoning and applying optimal token-zero curriculums natively during pre-training.

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Escaping the MCP Trap





The Industry Standard

Stuffing 50+ rigid tool calls into a system prompt. Forces the model to choose from predefined, static actions. High latency, fragile.

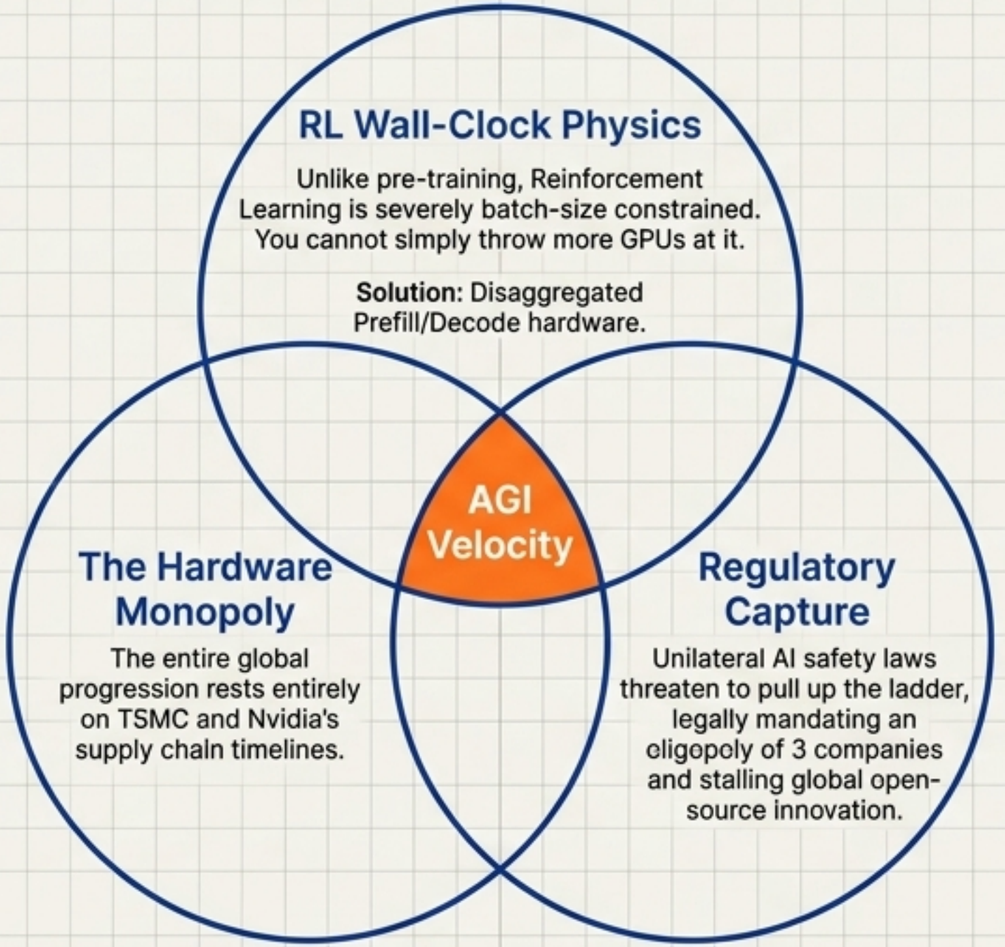
The Poolside Approach

Give the model a minimal free-coding container (Shell, Write, Fetch Web). The model dynamically writes Python scripts, uses for-loops, and interacts with APIs freely.

Agents don't want rigid tool calls; they want the freedom to write code to solve complex, long-horizon tasks.

NotebookLM

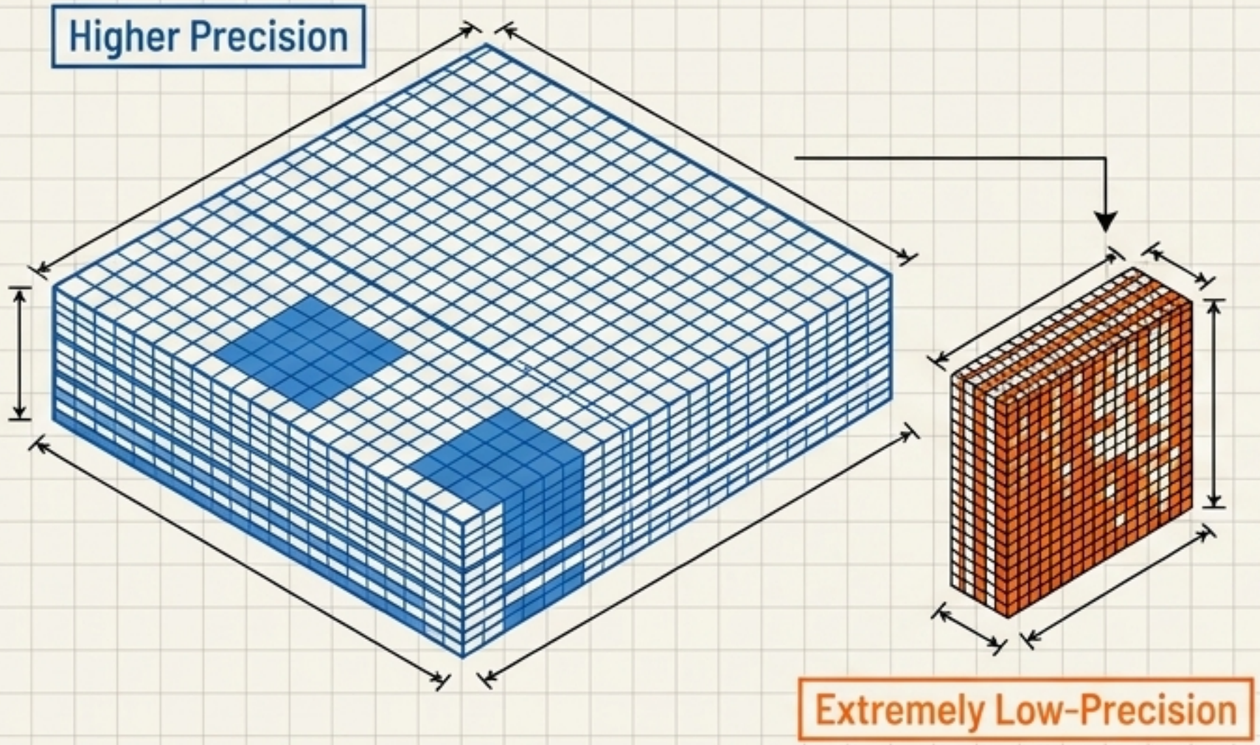
The True Bottlenecks to AGI



The existential threats to democratized AGI aren't math limits—they are hardware supply chain realities and premature regulatory lockdowns.

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Precision Over Scale



1. **A larger model at lower precision mathematically trumps a smaller model at higher precision.**

   Empirical data confirms computational efficiency gains outweigh potential accuracy losses from reduced bit-width, enabling training of vastly larger models.

2. **Current State: Laguna S trained in FP8.**

   Our current state-of-the-art models leverage FP8 mixed-precision training to maximize performance on existing hardware.

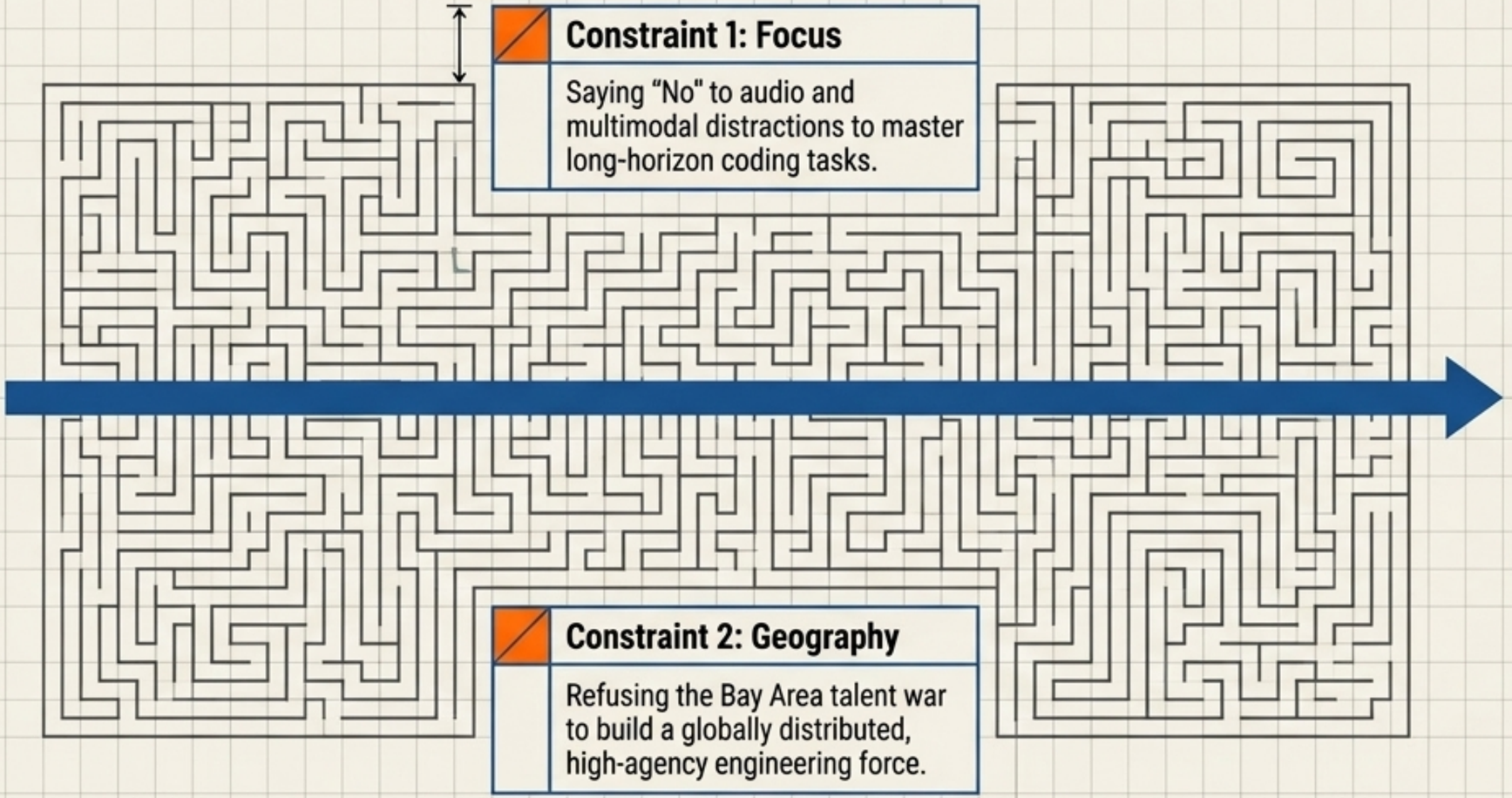
3. **The Next Frontier: Moving toward extremely low-precision training (NVFP4 and Ternary).**

   Future development is focused on harnessing extreme low-precision formats like NVFP4 and Ternary to break through current compute and memory walls.

Because our absolute bottleneck is MatMul compute and networking speed, aggressive precision reduction is the highest-leverage lever to squeeze frontier intelligence out of constrained clusters.

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Innovation Through Constraints



Constraint 1: Focus

Saying “No" to audio and multimodal distractions to master long-horizon coding tasks.

Constraint 2: Geography

Refusing the Bay Area talent war to build a globally distributed, high-agency engineering force.

The Poolside Mandate

High agency requires shared goals and absolute boundaries.

We are optimizing for individuals who want the highest ratio of personal impact to global outcome.

Never round down your ambitions. The 100-model ecosystem starts here.

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